INCH-POUND

MS51515B
17 March 2016
SUPERSEDING
MS51515A
17 January 1979

DETAIL SPECIFICATION SHEET

TEE, TUBE, BULKHEAD ON BRANCH, 37 DEGREE FLARED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design after 17 August 1999. For new design, use SAE-J514.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.

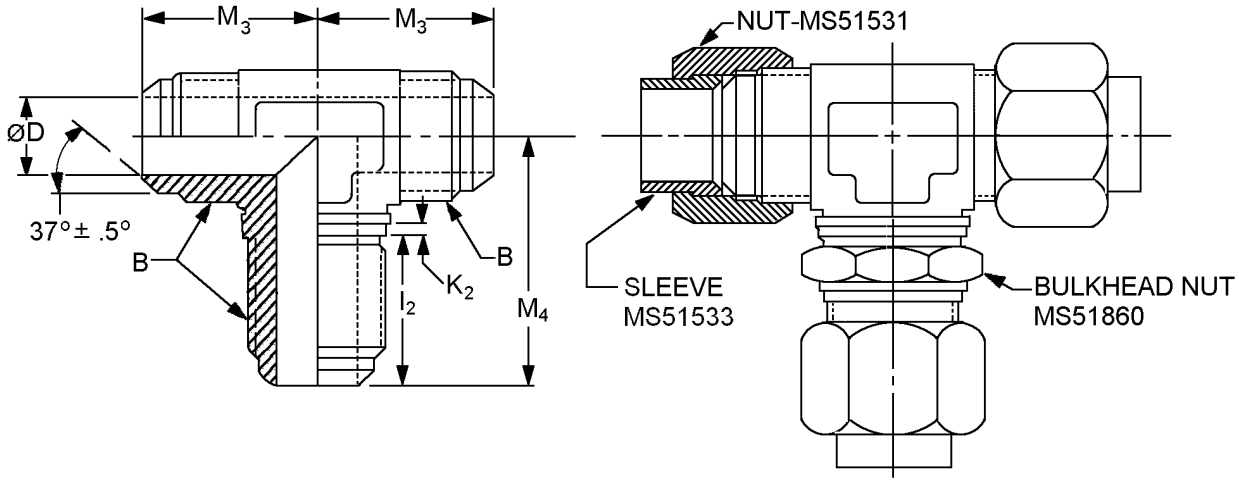


FIGURE 1. Tee, female pipe on run.

AMSC N/A

FSC 4730

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

Suffix designator		Tube OD nom.	B Straight thread	D diameter	
Assembly	Body			Basic inches (mm)	Tolerance inches (mm)
A2	B2	.1250 (3.175)	.3125-24 UNF-2A	.062 (1.57)	±.003 (0.08)
A3	B3	.1875 (4.763)	.3750-24 UNF-2A	.125 (3.18)	
A4	B4	.2500 (6.350)	.4375-20 UNF-2A	.172 (4.37)	
A5	B5	.3125 (7.936)	.5000-20 UNF-2A	.234 (5.94)	
A6	B6	.3750 (9.525)	.5625-18 UNF-2A	.297 (7.54)	±.004 (0.10)
A8	B8	.5000 (12.700)	.7500-16 UNF-2A	.391 (9.93)	
A10	B10	.6250 (15.875)	.8750-14 UNF-2A	.484 (12.29)	
A12	B12	.7500 (19.050)	1.0625-12 UN-2A	.609 (15.47)	±.005 (0.13)
A16	B16	.8750 (22.225)	1.3125-12 UN-2A	.844 (21.44)	±.007 (0.18)
A20	B20	1.0000 (25.400)	1.6250-12 UN-2A	1.078 (27.38)	+.008 -.005 (+0.20 -0.13)
A24	B24	1.2500 (31.750)	1.8750-12 UN-2A	1.312 (33.32)	
A32	B32	1.5000 (38.100)	2.500-12 UN-2A	1.781 (45.24)	+.010 -.005 (+0.25 -.013)

Suffix designator		l_2 inches (mm) ±.030 (0.76)	K_2 inches (mm) ±.020 (0.51)	M_3 inches (mm) ±.020 (0.51)	M_4 inches (mm) ±.030 (0.76)
Assembly	Body				
A2	B2	.920 (23.37)	.094 (2.39)	.840 (21.34)	1.420 (36.07)
A3	B3	.920 (23.37)	.094 (2.39)	.910 (23.11)	1.450 (36.83)
A4	B4	1.020 (25.91)	.094 (2.39)	.970 (24.64)	1.590 (40.39)
A5	B5	1.020 (25.91)	.094 (2.39)	1.030 (26.16)	1.620 (41.15)
A6	B6	1.090 (27.69)	.094 (2.39)	1.090 (27.69)	1.810 (40.89)
A8	B8	1.250 (31.75)	.125 (3.18)	1.360 (34.54)	2.110 (53.59)
A10	B10	1.390 (35.31)	.125 (3.18)	1.560 (39.62)	2.390 (60.71)
A12	B12	1.560 (39.62)	.125 (3.18)	1.780 (45.21)	2.670 (67.82)
A16	B16	1.560 (39.62)	.125 (3.18)	1.940 (49.28)	2.800 (71.12)
A20	B20	1.610 (40.89)	.125 (3.18)	2.170 (55.12)	3.120 (79.25)
A24	B24	1.620 (41.15)	.125 (3.18)	2.340 (59.44)	3.420 (86.87)
A32	B32	1.910 (48.51)	.125 (3.18)	2.890 (73.41)	4.110 (104.39)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges and remove all burrs and slivers.
4. Assemblies are only furnished to this specification (body, nuts, and sleeves). Bodies are not to be stocked stored or issued.
5. Dimensions and tolerances not shown shall be in accordance with SAE-J514 for 37° flared fittings.
6. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.

FIGURE 1. Tee, female pipe on run. - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and tables I and II.

Materials shall be in accordance with MIL-DTL-18866 and table I.

TABLE I. Materials.

Material	Form	Specification	Alloy
Carbon steel	Bar	SAE-J403	1117, 1126, 1140, 1213, 1215, or 12L14
	Forgings		
Chrome-molybdenum steel	Bars	SAE-AMS6370	4130
	Forgings	SAE-AMS6382	
		SAE-AMS6370	
Corrosion resistant steel	Bars and forgings	ASTM A276/A276M	304, 304L, 316, or 321
		ASTM A564/A564M	XM-12 (15-5 PH) UNS S15500 or 630 (17-4 PH) UNS S17400
		SAE-AMS5639	UNS S30400
		SAE-AMS5645	UNS S32100
		SAE-AMS5647	UNS S30403
		SAE-AMS5743	UNS S35500
	Bar	ASTM A582/A582M	UNS S30300
Nickel-copper alloy	Bar	ASTM B164 QQ-N-281	UNS N04400
High-chromium nickel alloy	Bar	ASTM B166	UNS N06690
	Forgings	ASTM B564	
Titanium 1/	Bars	SAE-AMS4928	6AI-4V annealed
	Forgings		

1/ Titanium shall not be used in oxygen or potable water systems.

Finish. Finishes shall be as specified in table II. All platings shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

For Carbon Steel: Alloys 1110 and 1120 though NOT listed in SAE J403 are said to be listed there by the UNS Guidebook - an inconsistency; there is NO 1120 listed in J403 or in the Guidebook - we recommend calling for alloys 1117 and 1126 in lieu of 1110 and 1120. These are supposed to be 'free-machining' alloys, and as such the suggested 'in lieu of' replacements follow that intention.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE II. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 2 or SAE-AMS-QQ-P-416, type II, class 2. 1/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC.
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 3/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 4. 2/
R	Steel	Zinc plating in accordance with ASTM B633; type VI, Fe/Zn 12. 5/
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 4/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 4/
Z	Steel	Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 12, or ASTM B695, type II, class 12. 5/
ZC	Steel	Zinc may be any zinc plating from PIN codes H, J, and R with a colored chromate coating 5/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems.
2/ Hexavalent chromium free. Finish shall be ROHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum coating thickness.
4/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to FED-STD-595 colors 36076 through 36293.
5/ Not for use in aircraft. Requires approval from the Program Officer for all applications.

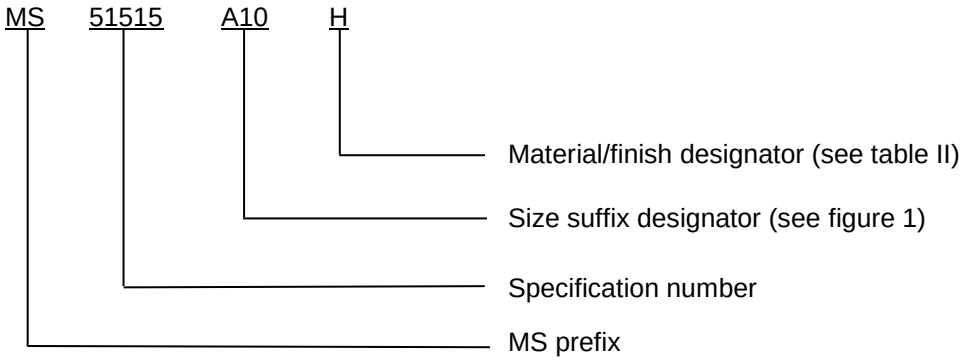
Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using one of the brush plating processes below as appropriate to primary finish. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

PIN: The PIN consists of the letters “MS”, the specification number, a letter and number for tee size, and a letter for material finish designator.



PIN example: MS51515A10H indicates a tee, pipe to tube, .6250 inch (15.875 mm), steel zinc-aluminum.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

The listing below provides a detailed cross-reference of inactive MS51515 PIN’s and for new design SAE-J514 PIN’s.

MS51515 parts have straight threads in accordance with ASME B1.1 the SAE parts have straight threads in accordance with SAE-J425.

Plating “P” SAE allows a range of nickel from 6% to 20%. Below 12%, ZnNi is not much better than zinc plating, which is less expensive and easier to apply. Above 16%, ZnNi becomes more cathodic and no longer acts as a sacrificial coating - if a high nickel coating is damaged the steel beneath the coating will corrode at an accelerated rate.

Users are cautioned to evaluate replacement parts for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE III. MS51515 to SAE-J514 cross reference.

Inactive for new design MS51515- PIN		TUBE O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51515A2	MS51515B2	1/8	Not available	Not available
MS51515A2H	MS51515B2H	1/8	Not available	Not available
MS51515A2J	MS51515B2J	1/8	Not available	Not available
MS51515A2M	MS51515B2M	1/8	J514-2-2-2-070959MA	J514-2-2-2-070959MB
MS51515A2N	MS51515B2N	1/8	J514-2-2-2-070959NA	J514-2-2-2-070959NB
MS51515A2P	MS51515B2P	1/8	J514-2-2-2-070959PA	J514-2-2-2-070959PB
MS51515A2	MS51515B2	1/8	J514-2-2-2-070959QA	J514-2-2-2-070959QB
MS51515A2S	MS51515B2S	1/8	J514-2-2-2-070959SA	J514-2-2-2-070959SB
MS51515A2T	MS51515B2T	1/8	J514-2-2-2-070959TA	J514-2-2-2-070959TB
MS51515A2TF	MS51515B2TF	1/8	Not available	Not available
MS51515A2Z	MS51515B2Z	1/8	J514-2-2-2-070959ZA	J514-2-2-2-070959ZB
MS51515A2ZC	MS51515B2ZC	1/8	Not available	Not available
MS51515A3	MS51515B3	3/16	Not available	Not available
MS51515A3H	MS51515B3H	3/16	Not available	Not available
MS51515A3J	MS51515B3J	3/16	Not available	Not available
MS51515A3M	MS51515B3M	3/16	J514-3-3-3-070959MA	J514-3-3-3-070959MB
MS51515A3N	MS51515B3N	3/16	J514-3-3-3-070959NA	J514-3-3-3-070959NB
MS51515A3P	MS51515B3P	3/16	J514-3-3-3-070959PA	J514-3-3-3-070959PB
MS51515A3R	MS51515B3R	3/16	Not available	Not available
MS51515A3S	MS51515B3S	3/16	J514-3-3-3-070959SA	J514-3-3-3-070959SB
MS51515A3T	MS51515B3T	3/16	J514-3-3-3-070959TA	J514-3-3-3-070959TB
MS51515A3TF	MS51515B3TF	3/16	Not available	Not available
MS51515A3Z	MS51515B3Z	3/16	J514-3-3-3-070959ZA	J514-3-3-3-070959ZB
MS51515A3ZC	MS51515B3ZC	3/16	Not available	Not available
MS51515A4	MS51515B4	1/4	Not available	Not available
MS51515A4H	MS51515B4H	1/4	Not available	Not available
MS51515A4J	MS51515B4J	1/4	Not available	Not available
MS51515A4M	MS51515B4M	1/4	J514-4-4-4-070959MA	J514-4-4-4-070959MB
MS51515A4N	MS51515B4N	1/4	J514-4-4-4-070959NA	J514-4-4-4-070959NB
MS51515A4P	MS51515B4P	1/4	J514-4-4-4-070959PA	J514-4-4-4-070959PB
MS51515A4R	MS51515B4R	1/4	Not available	Not available
MS51515A4S	MS51515B4S	1/4	J514-4-4-4-070959SA	J514-4-4-4-070959SB
MS51515A4T	MS51515B4T	1/4	J514-4-4-4-070959TA	J514-4-4-4-070959TB
MS51515A4TF	MS51515B4TF	1/4	Not available	Not available
MS51515A4Z	MS51515B4Z	1/4	J514-4-4-4-070959ZA	J514-4-4-4-070959ZB
MS51515A4ZC	MS51515B4ZC	1/4	Not available	Not available
MS51515A5	MS51515B5	5/16	Not available	Not available
MS51515A5H	MS51515B5H	5/16	Not available	Not available
MS51515A5J	MS51515B5J	5/16	Not available	Not available
MS51515A5M	MS51515B5M	5/16	J514-5-5-5-070959MA	J514-5-5-5-070959MB
MS51515A5N	MS51515B5N	5/16	J514-5-5-5-070959NA	J514-5-5-5-070959NB
MS51515A5P	MS51515B5P	5/16	J514-5-5-5-070959PA	J514-5-5-5-070959PB
MS51515A5	MS51515B5	5/16	Not available	Not available
MS51515A5S	MS51515B5S	5/16	J514-5-5-5-070959SA	J514-5-5-5-070959SB
MS51515A5T	MS51515B5T	5/16	J514-5-5-5-070959TA	J514-5-5-5-070959TB
MS51515A5TF	MS51515B5TF	5/16	Not available	Not available
MS51515A5Z	MS51515B5Z	5/16	J514-5-5-5-070959ZA	J514-5-5-5-070959ZB
MS51515A5ZC	MS51515B5ZC	5/16	Not available	Not available

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE III. MS51515 to SAE-J514 cross reference - Continued.

Inactive for new design MS51515- PIN		TUBE O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51515A6	MS51515B6	3/8	Not available	Not available
MS51515A6H	MS51515B6H	3/8	Not available	Not available
MS51515A6J	MS51515B6J	3/8	Not available	Not available
MS51515A6M	MS51515B6M	3/8	J514-6-6-6-070959MA	J514-6-6-6-070959MB
MS51515A6N	MS51515B6N	3/8	J514-6-6-6-070959NA	J514-6-6-6-070959NB
MS51515A6P	MS51515B6P	3/8	J514-6-6-6-070959PA	J514-6-6-6-070959PB
MS51515A6R	MS51515B6R	3/8	Not available	Not available
MS51515A6S	MS51515B6S	3/8	J514-6-6-6-070959SA	J514-6-6-6-070959SB
MS51515A6T	MS51515B6T	3/8	J514-6-6-6-070959TA	J514-6-6-6-070959TB
MS51515A6TF	MS51515B6TF	3/8	Not available	Not available
MS51515A6Z	MS51515B6Z	3/8	J514-6-6-6-070959ZA	J514-6-6-6-070959ZB
MS51515A6ZC	MS51515B6ZC	3/8	Not available	Not available
MS51515A8	MS51515B8	1/2	Not available	Not available
MS51515A8H	MS51515B8H	1/2	Not available	Not available
MS51515A8J	MS51515B8J	1/2	Not available	Not available
MS51515A8M	MS51515B8M	1/2	J514-8-8-8-070959MA	J514-8-8-8-070959MB
MS51515A8N	MS51515B8N	1/2	J514-8-8-8-070959NA	J514-8-8-8-070959NB
MS51515A8P	MS51515B8P	1/2	J514-8-8-8-070959PA	J514-8-8-8-070959PB
MS51515A8R	MS51515B8R	1/2	Not available	Not available
MS51515A8S	MS51515B8S	1/2	J514-8-8-8-070959PSA	J514-8-8-8-070959SB
MS51515A8T	MS51515B8T	1/2	J514-8-8-8-070959TA	J514-8-8-8-070959TB
MS51515A8TF	MS51515B8TF	1/2	Not available	Not available
MS51515A8Z	MS51515B8Z	1/2	J514-8-8-8-070959ZA	J514-8-8-8-070959ZB
MS51515A8ZC	MS51515B8ZC	1/2	Not available	Not available
MS51515A10	MS51515B10	5/8	Not available	Not available
MS51515A10H	MS51515B10H	5/8	Not available	Not available
MS51515A10J	MS51515B10J	5/8	Not available	Not available
MS51515A10M	MS51515B10M	5/8	J514-10-10-10-070959MA	J514-10-10-10-070959MB
MS51515A10N	MS51515B10N	5/8	J51410-10-10-070959NA	J514-10-10-10-070959NB
MS51515A10P	MS51515B10P	5/8	J514-10-10-10-070959PA	J514-10-10-10-070959PB
MS51515A10R	MS51515B10R	5/8	Not available	Not available
MS51515A10S	MS51515B10S	5/8	J514-10-10-10-070959SA	J514-10-10-10-070959SB
MS51515A10T	MS51515B10T	5/8	J514-10-10-10-070959TA	J514-10-10-10-070959TB
MS51515A10TF	MS51515B10TF	5/8	Not available	Not available
MS51515A10Z	MS51515B10Z	5/8	J514-10-10-10-070959ZA	J514-10-10-10-070959ZB
MS51515A10ZC	MS51515B10ZC	5/8	Not available	Not available

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE III. MS51515 to SAE-J514 cross reference - Continued.

Inactive for new design MS51515- PIN		TUBE O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51515A12	MS51515B12	3/4	Not available	Not available
MS51515A12H	MS51515B12H	3/4	Not available	Not available
MS51515A12J	MS51515B12J	3/4	Not available	Not available
MS51515A12M	MS51515B12M	3/4	J514-12-12-12-070959MA	J514-12-12-12-070959MB
MS51515A12N	MS51515B12N	3/4	J514-12-12-12-070959NA	J514-12-12-12-070959NB
MS51515A12P	MS51515B12P	3/4	J514-12-12-12-070959PA	J514-12-12-12-070959PB
MS51515A12R	MS51515B12R	3/4	Not available	Not available
MS51515A12S	MS51515B12S	3/4	J514-12-12-12-070959SA	J514-12-12-12-070959SB
MS51515A12T	MS51515B12T	3/4	J514-12-12-12-070959TA	J514-12-12-12-070959TB
MS51515A12TF	MS51515B12TF	3/4	Not available	Not available
MS51515A12Z	MS51515B12Z	3/4	J514-12-12-12-070959ZA	J514-12-12-12-070959ZB
MS51515A12ZC	MS51515B12ZC	3/4	Not available	Not available
MS51515A16	MS51515B16	1	Not available	Not available
MS51515A16H	MS51515B16H	1	Not available	Not available
MS51515A16J	MS51515B16J	1	Not available	Not available
MS51515A16M	MS51515B16M	1	J514-16-16-16-070959MA	J514-16-16-16-070959MB
MS51515A16N	MS51515B16N	1	J514-16-16-16-070959NA	J514-16-16-16-070959NB
MS51515A16P	MS51515B16P	1	J514-16-16-16-070959PA	J514-16-16-16-070959PB
MS51515A16R	MS51515B16R	1	Not available	Not available
MS51515A16S	MS51515B16S	1	J514-16-16-16-070959SA	J514-16-16-16-070959SB
MS51515A16T	MS51515B16T	1	J514-16-16-16-070959TA	J514-16-16-16-070959TB
MS51515A16TF	MS51515B16TF	1	Not available	Not available
MS51515A16Z	MS51515B16Z	1	J514-16-16-16-70959ZA	J514-16-16-16-070959ZB
MS51515A16ZC	MS51515B16ZC	1	Not available	Not available
MS51515A20	MS51515B20	1 1/4	Not available	Not available
MS51515A20H	MS51515B20H	1 1/4	Not available	Not available
MS51515A20J	MS51515B20J	1 1/4	Not available	Not available
MS51515A20M	MS51515B20M	1 1/4	J514-20-20-20-070959MA	J514-20-20-20-070959MB
MS51515A20N	MS51515B20N	1 1/4	J514-20-20-20-070959NA	J514-20-20-20-070959NB
MS51515A20P	MS51515B20P	1 1/4	J514-20-20-20-070959PA	J514-20-20-20-070959PB
MS51515A20R	MS51515B20R	1 1/4	Not available	Not available
MS51515A20S	MS51515B20S	1 1/4	J514-20-20-20-070959SA	J514-20-20-20-070959SB
MS51515A20T	MS51515B20T	1 1/4	J514-20-20-20-070959TA	J514-20-20-20-070959TB
MS51515A20TF	MS51515B20TF	1 1/4	Not available	Not available
MS51515A20Z	MS51515B20Z	1 1/4	J514-20-20-20-070959ZA	J514-20-20-20-070959ZB
MS51515A20ZC	MS51515B20ZC	1 1/4	Not available	Not available

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE III. MS51515 to SAE-J514 cross reference - Continued.

Inactive for new design MS51515- PIN		TUBE O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51515A24	MS51515B24	1 1/2	Not available	Not available
MS51515A24H	MS51515B24H	1 1/2	Not available	Not available
MS51515A24J	MS51515B24J	1 1/2	Not available	Not available
MS51515A24M	MS51515B24M	1 1/2	J514-24-24-24-070959MA	J514-24-24-24-070959MB
MS51515A24N	MS51515B24N	1 1/2	J514-24-24-24-070959NA	J514-24-24-24-070959NB
MS51515A24P	MS51515B24P	1 1/2	J514-24-24-24-070959PA	J514-24-24-24-070959PB
MS51515A24R	MS51515B24R	1 1/2	Not available	Not available
MS51515A24S	MS51515B24S	1 1/2	J514-24-24-24-070959SA	J514-24-24-24-070959SB
MS51515A24T	MS51515B24T	1 1/2	J514-24-24-24-070959TA	J514-24-24-24-070959TB
MS51515A24TF	MS51515B24TF	1 1/2	Not available	Not available
MS51515A24Z	MS51515B24Z	1 1/2	J514-24-24-24-070959ZA	J514-24-24-24-070959ZB
MS51515A24ZC	MS51515B24ZC	1 1/2	Not available	Not available
MS51515A32	MS51515B32	2	Not available	Not available
MS51515A32H	MS51515B32H	2	Not available	Not available
MS51515A32J	MS51515B32J	2	Not available	Not available
MS51515A32M	MS51515B32M	2	J514-2-32-32-070959MA	J514-32-32-32-070959MB
MS51515A32N	MS51515B32N	2	J514-32-32-32-070959NA	J514-32-32-32-070959NB
MS51515A32P	MS51515B32P	2	J514-32-32-32-070959PA	J514-32-32-32-070959PB
MS51515A32R	MS51515B32R	2	Not available	Not available
MS51515A32S	MS51515B32S	2	J514-32-32-32-070959SA	J514-32-32-32-070959SB
MS51515A32T	MS51515B32T	2	J514-32-32-32-070959TA	J514-32-32-32-070959TB
MS51515A32TF	MS51515B32TF	2	Not available	Not available
MS51515A32Z	MS51515B32Z	2	J514-32-32-32-070959ZA	J514-32-32-32-070959ZB
MS51515A32ZC	MS51515B32ZC	2	Not available	Not available

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

FED-STD-595/36076	FED-STD-595/36280	ASTM B164	SAE-AMS2488
FED-STD-595/36081	FED-STD-595/36293	ASTM B166	SAE-AMS2700
FED-STD-595/36099	MIL-DTL-16232	ASTM B564	SAE-AMS4928
FED-STD-595/36118	MS51531	ASTM B633	SAE-AMS5639
FED-STD-595/36134	MS51533	ASTM B695	SAE-AMS5645
FED-STD-595/36152	MS51860	ASTM F1136/F1136M	SAE-AMS5647
FED-STD-595/36170	QQ-N-281	SAE-AMS-C-81562	SAE-AMS5743
FED-STD-595/36173	ASME B1.1	SAE-AMS-QQ-P-416	SAE-AMS6370
FED-STD-595/36176	ASTM A276/A276M	SAE-AMS2417	SAE-AMS6382
FED-STD-595/36231	ASTM A564/A564M	SAE-AMS2451/5	SAE-J403
FED-STD-595/36251	ASTM A582/A582M	SAE-AMS2451/9	SAE-J425
FED-STD-595/36270	ASTM B117	SAE-AMS2486	SAE-J514

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2016-015)

Review activities:
Army - AT, MI
Navy - CG, MC, SA, SH
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.